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Feature-based attention modulates feedforward visual processing

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It is widely believed that attention selects locations at an earlier stage than it selects nonspatial features, but this has been tested only under conditions of minimal competition. We found that, when competition was increased, color-based attention was able to influence the feedforward flow of information in humans within 100 ms of stimulus onset, even for stimuli presented at unattended locations. Thus, color-based attention can operate as early as, and independently from, spatial attention.

Relevant information in the visual environment is often defined by nonspatial features, such as the shape of a face or the color of an apple. However, many researchers have argued that attentional selection of relevant information ultimately occurs on the basis of location[1,2], with nonspatial features being used to determine which locations should be attended. This has been clearest in studies of event-related potentials (ERPs), in which spatial attention enhances the amplitude of the sensory-evoked P1 component within 100 ms of stimulus onset, with feature-based attention effects typically occurring between 150 and 300 ms post-stimulus and probably reflecting feedback signals[3–5]. In addition, these feature-based attention effects are typically eliminated for stimuli presented at unattended locations[4,5]. These findings suggest that spatial attention precedes featural attention and that featural attention is applied only to stimuli selected by spatial attention.

However, ERP studies of featural attention have typically presented the attended and ignored feature values one at a time, minimizing direct competition between them. Attentional selection is strongly dependent on competition. Both spatial attention[6] and feature-based attention[7] effects may be much stronger when attended and ignored stimuli simultaneously compete for access to perceptual processing resources. Moreover, single-unit, functional imaging and steady-state ERP studies using simultaneous presentation of attended and ignored feature values have found featural attention effects in extrastriate visual cortex for stimuli that are presented at ignored locations[8–10]. However, these studies did not address the timing of the attentional modulation or the question of whether attention was influencing feedforward or feedback activity.

Using ERP recordings, we found that feature-based attention can influence feedforward sensory activity, as reflected by the P1 wave, under conditions of simultaneous competition between attended and ignored feature values. A continuous stream of intermixed red and green dots was presented in one visual field, and observers attended either to the red or green dots to detect occasional luminance decrements in the attended color (**Fig. 1a**,**b** and **Supplementary Methods** online). To probe the selectivity of the visual system for the attended and ignored colors, we flashed all-red or all-green probe arrays in the opposite visual field. The task was sufficiently difficult to require continuous attention to the task-relevant stimulus stream, but subjects were able to maintain central fixation (**Supplementary Results** online). If featural attention can influence feedforward sensory processing independently of spatial attention, then the task-irrelevant probes should elicit a larger P1 wave when presented in the attended color than when presented in the unattended color, even though they were presented at an unattended location. We tested this hypothesis at

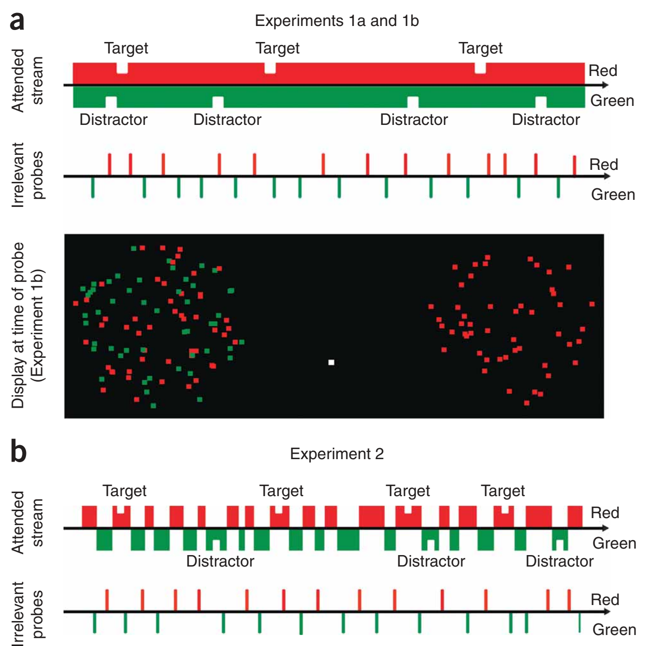

Figure 1 Experimental design. (**a**) Stimuli and timeline for the attend-red condition in Experiments 1a and 1b. The red and green dots were randomly intermingled throughout each 15-s trial, and subjects detected occasional luminance decrements (notches) in the attended color. Task-irrelevant all-red or all-green probe stimuli were flashed intermittently. The background was gray for Experiment 1a and black (as shown) for Experiment 1b. (**b**) Timeline for the attend-red condition in Experiment 2, in which the task-relevant stimuli were all-red or all-green at any given moment. Movies of the stimuli are presented in **Supplementary Videos 1–3** online.

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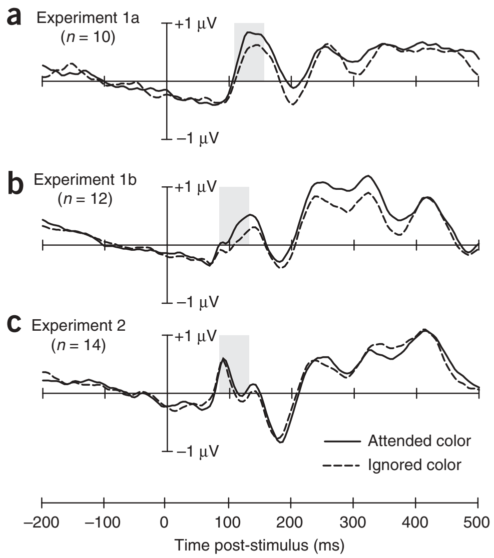

Figure 2 Grand average waveforms at contralateral occipital electrode sites. Waveforms are shown for attended-color (solid lines) and unattended-color probes (broken lines) for Experiment 1a (**a**), Experiment 1b (**b**) and Experiment 2 (**c**). Shaded areas indicated the ERP amplitude measurement windows.

different stimulus contrast levels in Experiments 1a and 1b and found that the timing of the attention effect depended on the timing of the stimulus-evoked sensory response.

As would be expected from prior research, sensory-evoked activity began approximately 30 ms earlier for the high-contrast experiment than for the low-contrast experiment (**Supplementary Results**). For both experiments, P1 amplitude over the cortex contralateral to the probe (**Fig. 2a**,**b**) was significantly larger for probes presented in the attended color than for probes presented in the unattended color (Experiment 1a, 110–160 ms, $P = 0.05$; Experiment 1b, 80–130 ms, $P = 0.02$; see **Supplementary Results** and **Supplementary Fig. 1** online for ipsilateral waveforms). This effect was evident within 100 ms of stimulus onset for the high-contrast stimuli used in Experiment 1b (80–100 ms, $P = 0.04$) and was observed across the lateral occipital electrode sites. Thus, color-based attention can influence the flow of feedforward sensory information within 100 ms of stimulus onset, even for stimuli presented at an unattended location.

Previous studies of color-based attention have not typically yielded P1 attention effects and, in Experiment 2, we tested the hypothesis that P1 amplitude was influenced by color-based attention in Experiments 1a and 1b because of the simultaneous competition between the attended and ignored colors. To test this hypothesis, the red and green elements of the task-relevant stimuli were presented sequentially, rather than simultaneously, in Experiment 2, and were thus never in direct competition (**Fig. 1b** and **Supplementary Methods**). We used high-contrast stimuli, as in Experiment 1b, and task difficulty was similar across experiments (**Supplementary Results**).

Color-based attention did not influence P1 amplitude in Experiment 2 (80–130 ms, $P = 0.23$; 80–100 ms, $P = 0.72$; **Fig. 2c**). Thus, in the absence of simultaneous competition, attention did not influence early feedforward sensory activity. A previous study using sequentially presented attended and ignored colors found an enhanced positivity from 100–140 ms for attended stimuli[11], and we found the same effect with approximately the same scalp distribution (100–140 ms, $P = 0.05$). The previous study concluded that this effect did not reflect a modulation of the sensory-evoked P1 wave, and the delayed onset of the effect relative to the onset of the P1 wave in the present experiment supports this conclusion.

These results provide, to the best of our knowledge, the first unambiguous demonstration that, under conditions of simultaneous competition between attended and ignored feature values, feature-based attention can influence feedforward sensory processing, even at an ignored location. In the domain of spatial attention, previous research has shown that task instructions lead to top-down signals from prefrontal and parietal cortex that create a tonic change in gain for the attended location in visual cortex[12] and that this leads to enhanced feedforward transmission when a stimulus appears at that location[4,6]. Top-down signals have also been shown for feature-based attention[13] and our results demonstrate that these signals lead to enhanced feedforward transmission for stimuli presented in the attended color, even when they are presented at an unattended location.

A previous study using simultaneous competition between attended and unattended colors found a similar attention effect for stimuli at an attended location[14], although it was not clear whether it was based on color or apparent depth. Considered together, our results and those of the previous study suggest that, under conditions of simultaneous competition, color-based attention operates throughout the visual field in a global manner (see **Supplementary Discussion** and **Supplementary Fig. 2** online), independently of spatial attention, as proposed by the similarity gain model[9,15]. Spatial attention may still be unique, however, because only spatial attention appears to influence early sensory processing in the absence of high levels of competition.

Note: Supplementary information is available on the Nature Neuroscience website.

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AUTHOR CONTRIBUTIONS
W.Z. and S.J.L. conceptualized and designed the experiments. W.Z. collected the data and performed the data analyses. W.Z. and S.J.L. wrote the paper.

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1. Treisman, A.M. & Gelade, G. *Cognit. Psychol.* **12**, 97–136 (1980).
2. Wolfe, J.M. *Psychon. Bull. Rev.* **1**, 202–238 (1994).
3. Eimer, M. *Biol. Psychol.* **41**, 167–182 (1995).
4. Hillyard, S.A. & Muente, T.F. *Percept. Psychophys.* **36**, 185–198 (1984).
5. Anllo-Vento, L. & Hillyard, S.A. *Percept. Psychophys.* **58**, 191–206 (1996).
6. Luck, S.J., Chelazzi, L., Hillyard, S.A. & Desimone, R. *J. Neurophysiol.* **77**, 24–42 (1997).
7. Saenz, M., Buracas, G.T. & Boynton, G.M. *Vision Res.* **43**, 629–637 (2003).
8. Saenz, M., Buracas, G.T. & Boynton, G.M. *Nat. Neurosci.* **5**, 631–632 (2002).
9. Treue, S. & Martinez Trujillo, J.C. *Nature* **399**, 575–579 (1999).
10. Muller, M.M. *et al. Proc. Natl. Acad. Sci. USA* **103**, 14250–14254 (2006).
11. Anllo-Vento, L., Luck, S.J. & Hillyard, S.A. *Hum. Brain Mapp.* **6**, 216–238 (1998).
12. Chawla, D., Rees, G. & Friston, K.J. *Nat. Neurosci.* **2**, 671–676 (1999).
13. Giesbrecht, B., Woldorff, M.G., Song, A.W. & Mangun, G.R. *Neuroimage* **19**, 496–512 (2003).
14. Valdes-Sosa, M., Bobes, M.A., Rodriguez, V. & Pinilla, T. *J. Cogn. Neurosci.* **10**, 137–151 (1998).
15. Boynton, G.M. *Curr. Opin. Neurobiol.* **15**, 465–469 (2005).